

PROJECT REPORT

CHRONIC DISEASE PREDICTION SYSTEM

Predicting Diabetes Mellitus Using Advanced Machine Learning Classification Algorithms

Course : Data Mining

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Dataset : Pima Indians Diabetes Dataset

1. Executive Summary

Early identification of individuals at high risk for chronic conditions like Diabetes Mellitus is critical for lowering long-term healthcare complications and designing targeted, proactive treatment workflows. This project develops a predictive diagnostic pipeline utilizing the foundational Pima Indians Diabetes Dataset. By leveraging predictive models, healthcare systems can transition from reactive disease management to early, automated diagnostic support tools.

The primary analytical contribution of this work lies in identifying and resolving critical biological data discrepancies—specifically, missing patient features mistakenly encoded as zero values (0)—and systematically benchmarking four prominent classification frameworks: Logistic Regression, Decision Trees, Random Forest, and Gradient Boosting. The evaluation relies on structural Key Performance Indicators (KPIs): Test Accuracy, Precision, Recall, F1-Score, and Area Under the Receiver Operating Characteristic curve (ROC-AUC).

Key Finding

Our optimized **Decision Tree Classifier** achieved the highest diagnostic accuracy of **78.57%** and an F1-Score of **69.16%**, presenting the highest equilibrium between identifying true positive cases and preventing false alarms. Concurrently, ensemble methodologies such as **Gradient Boosting** and **Random Forest** demonstrated superior generalization capabilities, yielding maximum ROC-AUC values of **81.87%** and **80.98%** respectively, asserting their long-term value for clinical risk stratification.

2. Problem Statement & Objectives

2.1 Clinical Motivation

Diabetes Mellitus is a major global contributor to chronic vascular, renal, and cardiovascular diseases. Traditional chemical screenings or symptomatic assessments catch the disease only after metabolic damage has advanced. Implementing automated classification metrics on fundamental vital signs and diagnostic tests allows clinicians to predict diabetic onset years before clinical manifestation.

2.2 Project Objectives

- **Automated Data Cleaning & Imputation:** Design an institutional preprocessing pipeline to accurately handle hidden missing values without causing data leakage.
- **Statistical Normalization:** Scale multi-dimensional physiological metrics (such as Plasma Glucose vs. Pedigree Function scales) to a standard distribution for stable learning.
- **Model Optimization & Cross-Tuning:** Train, regularize, and compare classic linear models against deep non-linear and ensemble structures.
- **Clinical Risk Simulation:** Validate the final models on newly simulated patient vitals to evaluate diagnostic utility.

3. Exploratory Data Analysis & Advanced Preprocessing

3.1 Dataset Overview

The system evaluates 768 patient records with 8 continuous features and 1 binary categorical targets. The target vector Outcome maps to 1 (Diabetic) or 0 (Healthy)

Feature Name	Measurement Unit	Clinical Metric Signification
Pregnancies	Count	Number of times pregnant
Glucose	mg/dL	2-hour plasma glucose concentration in an oral glucose tolerance test
BloodPressure	mm Hg	Diastolic blood pressure
SkinThickness	mm	Triceps skin fold thickness
Insulin	mu U/mL	2-hour serum insulin test result
BMI	Weight / Height ²	Body mass index score
DiabetespedigreeFunction	Index Score	Genetic diabetes history score matching familial risk curves
Age	Years	Patient chronological age

3.2 Biological Data Leakage Correction (Zero-Value Imputation)

A central data mining flaw identified in raw clinical sets is the encoding of missing parameters as a zero value (0). From a medical perspective, fields such as Blood Pressure, Plasma Glucose, or BMI cannot be zero in living individuals. Leaving these values unmodified introduces severe negative distribution skewness, completely misaligning the decision thresholds of linear estimators like Logistic Regression. To fix this, our pipeline isolates these false metrics and registers them to NaN representations. The calculated missing records across the features are summarized below:

- **Plasma Glucose:** 5 invalid zero-values.
- **Diastolic Blood Pressure:** 35 invalid zero-values.
- **Triceps Skin Fold Thickness:** 227 invalid zero-values (Extensive missingness).
- **Serum Insulin:** 374 invalid zero-values (High missingness rate at 48.6%).
- **Body Mass Index (BMI):** 11 invalid zero-values.

To avoid data leakage from the test set into training phases, the records are split into 80% train and 20% test matrices. We instantiate a scikit-learn SimpleImputer using a median strategy. The internal medians of the training partitions are calculated and imputed downstream across both partitions.

3.3 Feature Scaling & Transformation

Features exhibit highly disparate dimensional bands (e.g., Insulin values rise to multiple hundreds, while the Pedigree Function remains a fraction below 2.0). Feature standard deviation scales are stabilized using a StandardScaler , scaling each column vector to unit variance and zero mean using:

$$z = (x - \mu) / \sigma$$

This ensures distance-based metrics and gradient updates treat all elements identically during mathematical optimization.

4. Algorithmic Frameworks & Hyperparameter Optimization

Four structural classifiers were evaluated, moving from rigid statistical assumptions to highly scalable tree ensembles:

1. **Logistic Regression:** Utilized as our baseline linear estimator. Regularization penalty parameter adjusted to $C=0.5$ with the `max_iter` ceiling locked to 500 loops to ensure maximum optimization convergence without parameter oscillation.
2. **Decision Tree Classifier:** Structured to capture localized non-linear dependencies. To guarantee structural pruning and eliminate overfitting on shallow branches, parameters were locked to a max vertical depth of 4 levels (`max_depth=4`), forcing a minimum sample split barrier of 5 nodes (`min_samples_split=5`), and leaf boundaries fixed to `min_samples_leaf=2`.
3. **Random Forest Classifier:** An ensemble bagging model utilizing 200 distinct tree branches (`n_estimators=200`) trained parallelly over random data subsets. Vertical depth was constrained to 7 nodes to preserve structural generalization across multi-collinear inputs.
4. **Gradient Boosting Classifier (GBM):** A sequential boosting framework optimized via error-gradient updates. Tuned with 150 weak decision stumps (`n_estimators=150`) running at a smooth learning step threshold of 0.05 (`learning_rate=0.05`). Subsampling density was bounded to 80% (`subsample=0.8`) to integrate stochastic updates, driving resilient learning parameters against feature anomalies.

5 .Model Performance Metrics and Evaluation

Following strict validation execution across the identical testing sample matrix (154 patient observations), models were sorted and scored against five performance metrics:

Classification Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Decision Tree	0.7857	0.6981	0.6852	0.6916	0.7887
Random Forest	0.7532	0.6905	0.5370	0.6042	0.8098
Gradient Boosting	0.7532	0.6667	0.5926	0.6275	0.8187
Logistic Regression	0.6948	0.5745	0.5000	0.5347	0.8120

5.1 Deep Metric Diagnostics & Interpretation

- **The Diagnostic Trade-Off (Recall vs. Precision):** In diagnostic systems, Recall (Sensitivity) is paramount. Missed cases (False Negatives) carry severe consequences as undiagnosed patients are sent home without preventative care. The Decision Tree model yielded an impressive Recall of 68.52%, significantly reducing false negative outcomes compared to the Random Forest model (53.70%).
- **The F1-Score Equilibrium:** Because the dataset features an unbalanced outcome scale (more healthy subjects than diabetic subjects), total Accuracy alone can be misleading. The F1-Score provides a fairer metric, as it computes the harmonic mean of Precision and Recall. The Decision Tree leads this field conclusively at 69.16%.
- **ROC-AUC Discriminative Strength:** While the Decision Tree excels at default class label boundaries, Gradient Boosting achieved the highest ROC-AUC score of 81.87%, followed closely by Random Forest at 80.98%. This indicates that these ensemble models offer superior probability separation, giving clinicians finer control over adjusting the threshold settings depending on operational constraints.

6. Clinical Risk Simulation on New Case Profile

To benchmark model performance in a live environment, a simulated patient file representing a typical clinical screening case was fed into our preprocessing pipeline:

Patient Screening Record inputs: Pregnancies: 2 | Glucose: 140 mg/dL | Blood Pressure: 85 mm Hg | Skin Thickness: 30 mm | Insulin: 0 μ U/mL (Missing input) | BMI: 31.2 | Pedigree Score: 0.450 | Age: 35 years.

Our pipeline successfully handled the missing insulin level by imputing the median value, scaled the features, and generated the following diagnosis using the Random Forest classifier.

Pipeline Diagnostic Output Summary

Diagnostic Prediction: ✓ LOW RISK (Healthy)

Calculated Metabolic Probability: The model evaluated a **46.49%** continuous probability of diabetic status. Because this falls below our classification cutoff threshold (50%), the patient is categorized as low risk, preventing an unnecessary medical intervention while maintaining close monitoring.

7. Project Conclusion & Operational Recommendations

This project successfully demonstrates the utility of machine learning pipelines in clinical risk prediction. While linear models like Logistic Regression proved overly simple for this complex dataset (scoring an accuracy of just 69.48%), tree-based approaches captured non-linear physiological interactions effectively. For deployment, we recommend a hybrid architecture: use the Decision Tree Model for immediate diagnostic screening due to its high F1-Score and transparent rule path, while utilizing the Gradient Boosting Model for advanced clinical risk tracking, leveraging its superior 81.87% AUC discrimination capacity. Future development should integrate advanced missing data techniques like K-Nearest Neighbors (KNN) or MICE imputation, along with continuous feature expansion, to enhance model precision in fast-paced healthcare environments.